SIMATS ENGINEERING
ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics.* (BTL 5)

Assessment Tool Weightage: 10%

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

1. Detection Performance Comparison
Two object detection models produce the following results:

Model	True Positives	False Positives	False Negatives
A (Viola-Jones)	80	20	30
B (YOLO)	90	15	20

(a) Compute precision and recall for both models.
(b) Compare their performance and state which model is better for object detection.

1. Detection Performance Comparison

Formulas:

- Precision = TP / (TP + FP)
- Recall = TP / (TP + FN)

Model A – Viola-Jones

- TP = 80, FP = 20, FN = 30

- Precision = 80 / (80 + 20) = **0.80 = 80%**
- Recall = 80 / (80 + 30) = **0.7273 = 72.73%**

Model B – YOLO

- TP = 90, FP = 15, FN = 20
- Precision = 90 / (90 + 15) = **0.8571 = 85.71%**
- Recall = 90 / (90 + 20) = **0.8182 = 81.82%**

Model	**Precision**	**Recall**
Viola-Jones	80%	72.73%
YOLO	**85.71%**	**81.82%**

Conclusion: YOLO is better because it has both **higher precision and higher recall**. It produces fewer false positives and fewer false negatives, making it more effective for object detection.

2. Motion Estimation Error Analysis
Two motion estimation techniques produce displacement errors (in pixels):

Frame	**Method A**	**Method B**
1	**2**	**3**
2	**3**	**5**
3	**2**	**4**

(a) Compute average error for both methods.
(b) Compare and evaluate which method is more accurate.

Motion Estimation Error Analysis

Method A

Errors = 2, 3, 2

Average error:

Method B

Errors = 3, 5, 4

Average error:

Method	Average Error
Method A	2.33 pixels
Method B	4.00 pixels

Conclusion:

Method A is more accurate because its average displacement error is lower. A lower error indicates that the estimated motion is closer to the actual motion.

3. Optical Flow Magnitude Comparison
Optical flow magnitudes (in pixels/frame):

Region	Method A	Method B
R1	5	8
R2	6	10
R3	5	9

(a) Compute average motion magnitude for both methods.
(b) Compare and evaluate which method captures stronger motion.

Method A

Method B

Method	Average Motion Magnitude
Method A	5.33 pixels/frame
Method B	9.00 pixels/frame

Conclusion: Method B captures stronger motion because its average optical-flow magnitude is 9.00 pixels/frame, compared with 5.33 pixels/frame for Method A. This means Method B estimates larger motion vectors in the given regions.

4. Depth Estimation Comparison (Stereo vs SfM)
Estimated depth values (in meters):

Object	Stereo Vision	Structure from Motion
A	2.0	2.5
B	3.0	3.8
C	4.0	5.0

Actual depths: A=2, B=3, C=4
(a) Compute absolute error for both methods.
(b) Compare and evaluate which method is more accurate.

4. Depth Estimation Comparison

Absolute error:

Stereo Vision

- Object A: $|2.0 - 2.0| = 0.0$ m
- Object B: $|3.0 - 3.0| = 0.0$ m

- Object C: |4.0 − 4.0| = 0.0 m

Average absolute error:

Structure from Motion

- Object A: |2.5 − 2.0| = 0.5 m
- Object B: |3.8 − 3.0| = 0.8 m
- Object C: |5.0 − 4.0| = 1.0 m

Average absolute error:

Method	A Error	B Error	C Error	Average Error
Stereo Vision	0.0 m	0.0 m	0.0 m	0.00 m
SfM	0.5 m	0.8 m	1.0 m	0.77 m

Conclusion:

Stereo Vision is more accurate because its average absolute error is 0.00 m, while SfM has an average error of 0.77 m. Therefore, Stereo Vision gives the exact depth values for this dataset.

5. Pattern Recognition Accuracy Comparison

Two recognition systems produce results:

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

(a) Compute accuracy for both systems.

(b) Compare and evaluate which system performs better.

5. Pattern Recognition Accuracy Comparison

Traditional System

Deep Learning System

System	Correct Predictions	Accuracy
Traditional	85/100	85%
Deep Learning	92/100	**92%**

Conclusion:

Deep Learning performs better, with an accuracy of 92%, which is 7 percentage points higher than the traditional system's 85%.

Code of Conduct:

I <u>M Jeevitha (192424257)</u> (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Jeevitha

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with	Shows reasonable interpretation	Basic interpretation; lacks depth	No meaningful interpretation

		clear interpretation and reasoning	with minor gaps		
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand